

Problem Set 6 (two pages)

Statistics 24510-30040 (W25)

Due 9 am Tuesday, February 25th.

Requirements

- Upload your pdf named "Lastname-Firstname-hw6.pdf" to Gradescope under Pset-6. Be sure to mark the pages corresponding to each problem.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (Relevant reference: Rice chapter 14, Casella and Berger chapters 11, 12.)

1. (Simple derivations of linear models)

- (a) Suppose the model is $Y_i = \mu + \epsilon_i, i = 1, \dots, n$, ϵ_i are independent with mean zero and variance σ^2 . Derive the least square estimate of μ .
- (b) The linear model $Y_i = \beta x_i + \epsilon_i$ with ϵ_i i.i.d. mean zero is fitted to points $(x_i, y_i), i = 1, \dots, n$. Derive the least squares estimate of β .
- (c) Fitting a linear model $Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$ on n data points yields estimates $\hat{Y}$, $\hat{\beta}_1$, and $\hat{\beta}_2$. Now suppose each x_{1i} is replaced by $x_{1i}^* = kx_{1i}, i = 1, \dots, n$.
- Does $\hat{Y}$ change? If so, what's the new estimate?
 - Does $\hat{\beta}_1$ change? If so, what's the new estimate?
 - Does $\hat{\beta}_2$ change? If so, what's the new estimate?

2. Consider a linear model $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, with i.i.d. $\epsilon_i \sim N(0, \sigma^2)$ for $i = 1, 2, 3, 4$.

Before you collect the four data points, you have the freedom to choose x_1, x_2, x_3, x_4 and then observe y_1, y_2, y_3, y_4 . Suppose you can only choose x_1, x_2, x_3, x_4 in the range $[-2, 2]$.

How would you choose x_1, x_2, x_3, x_4 so that the accuracy of the MLE $\hat{\beta}_1$ is the best?

3. (MSE comparison)

The Mean Square Error $MSE(\hat{\theta})$ of an estimator $\hat{\theta}$ for a parameter θ is defined as $E(\hat{\theta} - \theta)^2$.

Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ random variables, $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean.

Consider two estimators of σ^2 :

Sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and the MLE $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$.

Compute and compare $MSE(s^2)$ and $MSE(\hat{\sigma}^2)$.

4. (Linear model design matrix and estimators)

Consider a standard linear regression model in which the freshman GPA (denoted by Y) is modeled to depend linearly on high school GPA (denoted by x). Consider the model

$$Y_i = I_F(i)\beta_F + I_M(i)\beta_M + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n \quad (1)$$

where

$$I_F(i) = \begin{cases} 1, & \text{if student } i \text{ is female,} \\ 0, & \text{if otherwise.} \end{cases}, \quad I_M(i) = \begin{cases} 1, & \text{if student } i \text{ is male,} \\ 0, & \text{if otherwise.} \end{cases}$$

Assuming that everyone is classified into M or F for this question.

- (a) Give the form of the design matrix X for the model.
- (b) Find $X^T X$. Simplify the matrix using n_f, n_m , the number of female and male freshman, respectively.

5. (*Model misspecification*) (related to the previous question)

As in the previous question, consider a linear regression model in which the freshman GPA (denoted as Y) is modeled to depend linearly on high school GPA (denoted as x).

Suppose at a particular college that the true $\beta_1 = 0$ (in the model (1) in the previous question), so that the true data-generating mechanism is

$$Y_i = I_F(i)\beta_F + I_M(i)\beta_M + e_i, \quad e_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n. \quad (2)$$

In addition, suppose that girls at this college tend to have higher GPA (denoted by x) than boys in both high school and freshman year of college. Now suppose the college (without knowing the truth) fits the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + e_i, \quad e_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n, \quad (3)$$

to a representative sample of students from this college (with roughly equal number of male and female students in the sample).

- (a) Find an exact expression for $\mathbb{E}(\hat{\beta}_1)$ obtained by fitting model (3) by least squares to the data when in fact model (2) is true.
- (b) Will this expected value in (a) generally be positive, negative or (nearly) 0? Explain.
- (c) Comment on what your finding says about how using model (3) rather than the correct (2) could result in misleading conclusions about the effect of high school GPA on college GPA.

6. (*Pythagorean identities*) Review and derive the following Pythagorean identities.

- (a) For any random variable $\hat{\theta}$ and any scalar $\theta \in \mathbb{R}$,

$$\mathbb{E}[(\hat{\theta} - \theta)^2] = \text{var}(\hat{\theta}) + [\mathbb{E}(\hat{\theta}) - \theta]^2$$

- (b) For any real numbers $x_1, \dots, x_n, \theta \in \mathbb{R}$,

$$\sum_{i=1}^n (x_i - \theta)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \theta)^2$$

where $\bar{x} = \sum_{i=1}^n x_i / n$.

- (c) For any vectors $v_1, \dots, v_n, \gamma \in \mathbb{R}^p$,

$$\sum_{i=1}^n \|v_i - \gamma\|^2 = \sum_{i=1}^n \|v_i - \bar{v}\|^2 + n\|\bar{v} - \gamma\|^2,$$

where $\bar{v} = \sum_{i=1}^n v_i / n$, and $\|\cdot\|$ the Euclidean norm for vectors, $\|w\|^2 = w^T w = \sum_{j=1}^p w_j^2$ for $w = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p$.

- (d) For any full-rank matrix $X \in \mathbb{R}^{n \times p}$, vectors $y \in \mathbb{R}^{n \times 1}$, $\beta \in \mathbb{R}^{p \times 1}$, where $n > p$,

$$\|y - X\beta\|^2 = \|y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2$$

where $\hat{\beta} = (X^T X)^{-1} X^T y$.